

Research Report 03:

Temporal Segmentation Strategy

Linguistically-Motivated 5-Second Clip Duration

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Abstract

This report documents the research, analysis, and decision-making process behind the 5-second temporal segmentation strategy for sign language video clips in the Voice-of-Hands dataset. We present empirical analysis of sign phrase durations, interpreter translation lag measurements, computational considerations, and comparison with existing datasets. The 5-second window with 50% overlap emerged as the optimal balance between linguistic completeness, model training efficiency, and dataset size.

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1 Introduction

1.1 Problem Statement

After detecting and extracting active signing segments, the system must partition the continuous video stream into discrete clips suitable for:

1. Sign language recognition (SLR) model training
2. Sign language translation (SLT) model training
3. Audio-visual alignment
4. Human review and annotation

The clip duration directly impacts:

- **Linguistic completeness:** Clips must contain sufficient context for comprehension
- **Computational efficiency:** Shorter clips enable faster training
- **Dataset size:** Longer clips reduce total clip count from fixed video duration
- **Alignment quality:** Duration must accommodate interpreter translation lag

1.2 Research Questions

1. What is the typical duration of sign language units (signs, phrases, sentences)?
2. What translation lag exists between spoken speech and sign interpretation?
3. How do existing datasets segment sign language video?
4. What clip duration optimizes the trade-off between context and efficiency?
5. Should clips overlap, and if so, by how much?

2 Linguistic Analysis of Sign Durations

2.1 Sign Language Unit Hierarchy

Sign languages exhibit hierarchical linguistic structure:

1. **Individual Sign (Lexical Unit):** Single sign representing a concept
2. **Sign Phrase:** 2–4 signs forming a predicate or argument
3. **Sign Clause:** Complete proposition with subject, predicate, object
4. **Sign Sentence:** One or more clauses forming a complete statement
5. **Sign Discourse:** Multiple sentences forming a narrative

2.2 Empirical Duration Measurements

We manually annotated 200 sign units from 10 parliamentary session clips to measure durations:

Key Observations:

- 95% of sign phrases complete within 5.4 seconds
- 80% of sign clauses complete within 6.4 seconds
- Sign sentences frequently exceed 10 seconds

Table 1: Sign Unit Duration Distribution (N=200 samples)

Unit Type	Mean	Std Dev	Min	Max
Single sign	2.1s	0.8s	1.2s	4.5s
Sign phrase	3.2s	1.1s	2.0s	6.8s
Sign clause	4.8s	1.6s	2.5s	9.2s
Sign sentence	7.3s	2.9s	3.8s	15.4s

2.3 Sign Phrase as Target Unit

Rationale: Sign phrases represent the optimal granularity for dataset clips because:

1. **Linguistically meaningful:** Phrases convey complete semantic units (predicates, arguments)
2. **Self-contained:** Can be understood independently with minimal context
3. **Training suitability:** Phrase-level clips align with current SLR model architectures
4. **Manageable duration:** 2–6 seconds fits memory constraints of video models

Decision: Target clip duration should capture 1–2 complete sign phrases.

3 Interpreter Translation Lag Analysis

3.1 The Décalage Phenomenon

Sign language interpreters in live settings exhibit systematic delay between hearing spoken language and producing sign translation. This "décalage" arises from:

1. **Processing time:** Understanding spoken utterance
2. **Structural reordering:** Converting SOV to sign language grammar
3. **Lexical search:** Retrieving appropriate signs
4. **Articulatory planning:** Coordinating hand and body movements

3.2 Lag Measurement Methodology

We measured lag in 50 parliamentary clips using the following procedure:

1. Manually transcribe spoken audio with precise timestamps
2. Manually annotate corresponding sign production timestamps
3. Compute temporal offset: $\Delta = t_{\text{sign}} - t_{\text{speech}}$
4. Analyze distribution of Δ across samples

3.3 Results

Interpretation:

- Median lag of 3.15s means sign interpretation trails speech by ~ 3 seconds
- High variability (SD = 1.28s) reflects sentence complexity and interpreter style
- Maximum lag of 7.4s observed during complex legal terminology

Table 2: Interpreter Translation Lag (N=50 speech-sign pairs)

Statistic	Value
Mean lag	3.36 seconds
Median lag	3.15 seconds
Standard deviation	1.28 seconds
90th percentile	5.2 seconds
Minimum	1.8 seconds
Maximum	7.4 seconds

3.4 Implications for Clip Duration

For proper audio-visual alignment:

$$\text{Audio}[t, t + T] \leftrightarrow \text{Video}[t + \Delta, t + \Delta + T] \quad (1)$$

where T is clip duration and $\Delta \approx 3.36\text{s}$ is the lag.

Consideration: A 5-second audio clip starting at time t corresponds to sign video produced during $[t + 3.36, t + 8.36]$. To capture this fully:

- Extract audio: $[t, t + 5]$
- Extract video: $[t + 3, t + 8]$ (approximately)

However, in the current implementation, we extract video and audio from the same time window, deferring lag compensation to future work. This creates imperfect but usable alignment, as:

- 5-second window still captures partial overlap of speech-sign content
- Future iterations can refine alignment using word-level timestamps

4 Comparison with Existing Datasets

4.1 Survey of Dataset Segmentation Strategies

Table 3: Temporal Segmentation in Major Sign Language Datasets

Dataset	Language	Avg Duration	Segmentation Method
PHOENIX-2014T	DGS	7.1s	Sentence boundaries
How2Sign	ASL	5.8s	Subtitle timestamps
CSL-Daily	Chinese SL	4.2s	Manual annotation
BOBSL	BSL	Variable	Subtitle alignment
OpenASL	ASL	6.5s	Subtitle boundaries
Voice-of-Hands	SSL	5.0s	Fixed + overlap

Observations:

- Most datasets use 4–7 second clips
- Subtitle-based segmentation leads to variable duration (2–15s)
- Fixed-duration clips simplify batch processing
- 5 seconds is near the median of existing approaches

5 Clip Duration Exploration

5.1 Candidate Durations Evaluated

We evaluated six candidate durations: 3s, 4s, 5s, 6s, 8s, 10s.

5.2 Evaluation Criteria

1. **Phrase Coverage:** Percentage of clips containing ≥ 1 complete phrase
2. **Context Sufficiency:** Percentage containing sufficient context for understanding
3. **Computational Cost:** Memory and processing time for model training
4. **Dataset Yield:** Number of clips generated from 1 hour of video
5. **Alignment Quality:** Ability to accommodate interpreter lag

5.3 Comparison

Table 4: Clip Duration Comparison

Duration	3s	4s	5s	6s	8s	10s
Phrase coverage	68%	82%	92%	95%	98%	99%
Context sufficient	55%	75%	88%	92%	96%	97%
Clips per hour	1200	900	720	600	450	360
Memory (GB)	2.1	2.8	3.5	4.2	5.6	7.0
Training time	1.0×	1.3×	1.7×	2.0×	2.7×	3.3×
Lag accommodation	Poor	Fair	Good	Good	Excellent	Excellent

5.4 Trade-Off Analysis

5.4.1 3–4 Second Clips

Advantages:

- High clip count (more training samples)
- Low memory footprint
- Faster training iterations

Disadvantages:

- 32% of phrases incomplete (3s) or 18% (4s)
- Insufficient context for complex signs
- Does not accommodate interpreter lag ($\Delta = 3.36\text{s}$)
- Frequently splits phrases mid-execution

5.4.2 8–10 Second Clips

Advantages:

- Nearly 100% phrase coverage
- Excellent context
- Captures full sentences
- Fully accommodates lag with margin

Disadvantages:

- Lower clip count (360–450 per hour vs. 720 at 5s)
- High memory requirements (5.6–7.0 GB vs. 3.5 GB)
- Slower training ($2.7\text{--}3.3\times$ vs. $1.7\times$)
- Redundant content (multiple pauses within clip)

5.4.3 5–6 Second Clips (Optimal Range)

5 Seconds Selected:

- **Phrase coverage:** 92% (acceptably high)
- **Context:** 88% sufficient (good)
- **Clip yield:** 720 per hour (good sample count)
- **Memory:** 3.5 GB (manageable on consumer GPUs)
- **Lag accommodation:** Covers mean lag (3.36s) with margin
- **Alignment:** Matches existing dataset norms (4–7s)

6 Overlap Strategy

6.1 Motivation for Overlap

Overlapping clips provide:

1. **Data augmentation:** Increases effective dataset size
2. **Temporal diversity:** Same sign appears in different contexts
3. **Boundary robustness:** Phrases split by non-overlapping clips appear complete in overlapped versions
4. **Training stability:** More samples reduce overfitting

Table 5: Overlap Strategy Comparison (5s clips, 1 hour video)

Overlap	Clips Generated	Phrase Coverage	Redundancy
0% (no overlap)	720	92%	0%
25% (1.25s)	960	95%	25%
50% (2.5s)	1440	97%	50%
75% (3.75s)	2880	98%	75%

6.2 Overlap Percentages Evaluated

6.3 50% Overlap Selected

Rationale:

- **2× dataset size:** 1440 clips vs. 720 (no overlap)
- **Phrase coverage improvement:** 97% vs. 92%
- **Moderate redundancy:** 50% is standard in sliding window approaches
- **Boundary recovery:** Phrases split by clip boundaries appear complete in overlapped clip
- **Computational balance:** 2× data without excessive redundancy (75% overlap is wasteful)

6.4 Implementation

```
clip_duration = 5.0 # seconds
overlap_percent = 0.5
step_size = clip_duration * (1 - overlap_percent) # 2.5 seconds

num_clips = int((total_duration - clip_duration) / step_size) + 1

for i in range(num_clips):
    start_time = i * step_size
    end_time = start_time + clip_duration

    if end_time > total_duration:
        break

    extract_clip(video, start_time, end_time)
```

Example: 30-second active segment

- **No overlap:** 6 clips ([0, 5], [5, 10], [10, 15], [15, 20], [20, 25], [25, 30])
- **50% overlap:** 11 clips ([0, 5], [2.5, 7.5], [5, 10], [7.5, 12.5], ..., [25, 30])

7 Resolution Standardization

7.1 Spatial Resolution Selection

Detected PiP regions vary in size:

- Typical dimensions: 280×320 to 420×480 pixels

- Aspect ratio: approximately 1:1 (square) to 4:3

Standardization required for:

- Batch processing (fixed tensor dimensions)
- Model architecture input consistency
- Storage optimization

7.2 Candidate Resolutions

Table 6: Resolution Options

Resolution	File Size	Detail	Speed	Use Case
128×128	1.2 MB	Poor	Fast	Low-end models
256×256	4.8 MB	Good	Moderate	Balanced
512×512	19.2 MB	Excellent	Slow	High-quality

256×256 Selected:

- **Hand detail preservation:** Fingers visible, finger spelling recognizable
- **Facial expression capture:** Non-manual markers (eyebrows, mouth) discernible
- **Storage efficiency:** 4.8 MB per 5s clip (manageable for large datasets)
- **Model compatibility:** Standard input for I3D, SlowFast, C3D architectures
- **GPU memory:** Fits 16–32 clips per batch on consumer GPUs (11–16 GB VRAM)

7.3 Interpolation Method

```
resized = cv2.resize(cropped_frame, (256, 256),
                    interpolation=cv2.INTER_CUBIC)
```

INTER_CUBIC chosen over INTER.LINEAR for:

- Smoother gradients (important for hand edges)
- Better quality for upscaling small PiP regions
- Minimal computational overhead (<5% slower than LINEAR)

8 Frame Rate Considerations

8.1 Source Video: 30 FPS

Sri Lankan parliamentary broadcasts recorded at 30 FPS.

Table 7: Frame Rate Impact

FPS	Frames/5s	File Size	Motion Capture
10	50	1.6 MB	Poor (jerky)
15	75	2.4 MB	Fair
25	125	4.0 MB	Good
30	150	4.8 MB	Excellent

8.2 Downsampling Analysis

Evaluated frame rate reduction for storage/computational efficiency:

30 FPS Retained:

- Sign language articulation occurs at 10–15 Hz (hand movement frequency)
- Nyquist criterion requires ≥ 20 –30 FPS for faithful capture
- Downsampling to 15 FPS loses rapid finger movements
- Storage cost difference (4.8 MB vs. 2.4 MB) acceptable

9 Video Encoding

9.1 Codec Selection

```
fourcc = cv2.VideoWriter_fourcc(*'mp4v') # MPEG-4 codec
out = cv2.VideoWriter(output_file, fourcc, fps, (256, 256))
```

H.264 (mp4v) Selected:

- Industry standard, universal compatibility
- Efficient compression (4.8 MB vs. 36 MB uncompressed)
- Hardware acceleration available (GPU encoding/decoding)
- Preserves quality at moderate bitrates

9.2 Quality Settings

```
ffmpeg -i input.mp4 -c:v libx264 -crf 23 -preset medium output.mp4
```

- **CRF 23:** Constant rate factor (visually lossless)
- **Preset medium:** Balance encoding speed and compression

10 Problems Encountered and Solutions

10.1 Problem 1: Phrase Boundary Detection

Initial Approach: Attempted automatic phrase boundary detection using:

- Motion energy peaks and valleys
- Two-hand symmetry events

- Facial expression changes

Problems:

- Phrase boundaries highly ambiguous (inter-annotator agreement only 67%)
- Detection accuracy only 58%
- Variable-length clips complicated batch processing
- Development time >> benefit

Solution: Abandon automatic phrase boundary detection, use fixed 5s duration with overlap

- Simpler implementation
- Higher reliability
- Overlap strategy recovers split phrases

10.2 Problem 2: Lag Compensation Implementation

Challenge: Implementing proper lag offset requires:

- Per-phrase lag estimation (varies by complexity)
- Word-level timestamp alignment
- ASR output post-processing

Current Status: Deferred to future work

- Current alignment: Extract video and audio from same time window
- Future enhancement: Shift video window by estimated $\Delta = 3.36s$
- Requires validation on substantial dataset first

10.3 Problem 3: Border Artifacts in Resized Clips

Issue: Some PiP overlays had partial border frames included in crop, leading to white/gray bars on edges after 256×256 resize.

Solution:

```
# Tighten bounding box by 5% margin
margin = 0.05
w, h = bbox[2] - bbox[0], bbox[3] - bbox[1]
bbox_tight = [
    bbox[0] + w * margin,
    bbox[1] + h * margin,
    bbox[2] - w * margin,
    bbox[3] - h * margin
]
```

Reduced border artifacts in 95% of clips.

Table 8: Generated Dataset Statistics (61-minute session)

Metric	Value
Source video duration	61 minutes
Active signing time	38 minutes (62%)
Total clips generated (50% overlap)	732 clips
Total dataset duration	61 minutes
Clips per hour	720
Average file size	4.8 MB
Total video size	3.51 GB

Table 9: Phrase Completeness (N=100 clips)

Completeness	Count	Percentage
≥ 1 complete phrase	92	92%
Partial phrase only	8	8%
≥ 2 complete phrases	68	68%
≥ 3 complete phrases	24	24%

11 Experimental Validation

11.1 Dataset Statistics with 5s Clips

11.2 Phrase Coverage Validation

Manual review of 100 randomly sampled clips:

Conclusion: 5-second duration achieves 92% complete phrase coverage, validating the linguistically-motivated selection.

12 Conclusion

12.1 Summary

The 5-second clip duration with 50% overlap represents an optimal balance across multiple criteria:

- **Linguistic completeness:** 92% clips contain ≥ 1 complete phrase
- **Dataset size:** 720 clips per hour (1440 with overlap)
- **Computational efficiency:** 3.5 GB memory, $1.7\times$ training time
- **Alignment compatibility:** Accommodates mean 3.36s interpreter lag
- **Industry alignment:** Matches 4–7s range of existing datasets

12.2 Key Design Decisions

1. **Fixed duration:** 5.0 seconds (not variable)
2. **Overlap:** 50% (2.5-second step)
3. **Resolution:** 256×256 pixels

4. **Frame rate:** 30 FPS (preserved from source)
5. **Codec:** H.264 (mp4v)
6. **Target unit:** Sign phrase (2–4 signs)

12.3 Future Refinements

1. Implement lag compensation (offset video by $\Delta = 3.36\text{s}$)
2. Experiment with variable-length clips based on natural pause detection
3. Explore 6-second duration for improved sentence coverage
4. Develop phrase boundary detection for linguistic segmentation
5. Test cross-dataset generalization with different clip durations